# Advanced Trigonometry

## Section 1: Angle Sum, Difference & Double-Angle Formulas

**1** If $\sin\theta = \frac{3}{5}$ and $\theta$ is in Quadrant II, what is the exact value of $\sin(2\theta)$?

**a.** $-\frac{24}{25}$

**b.** $\frac{24}{25}$

**c.** $-\frac{7}{25}$

**d.** $\frac{7}{25}$

**2** What is the exact value of $\cos(75^\circ)$ using the angle sum formula $\cos(45^\circ + 30^\circ)$?

**a.** $\frac{\sqrt{6}-\sqrt{2}}{4}$

**b.** $\frac{\sqrt{6}+\sqrt{2}}{4}$

**c.** $\frac{\sqrt{2}-\sqrt{6}}{4}$

**d.** $\frac{\sqrt{3}-1}{2}$

**3** Which of the following is equivalent to $\frac{1-\cos(2x)}{2}$?

**a.** $\sin^2 x$

**b.** $\cos^2 x$

**c.** $\tan^2 x$

**d.** $\sin(2x)$

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## Section 2: Solving Trigonometric Equations

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**4** Solve the equation $2\sin^2 x - 3\sin x + 1 = 0$ for all solutions in the interval $[0, 2\pi)$.

**a.** $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{\pi}{2}$

**b.** $\frac{\pi}{3}, \frac{2\pi}{3}, \pi$

**c.** $\frac{\pi}{6}, \frac{7\pi}{6}$

**d.** $\frac{\pi}{2}, \frac{3\pi}{2}$

**5** How many solutions does the equation $\cos(2x) = 0$ have in the closed interval $[0, 2\pi]$?

**a.** 2

**b.** 4

**c.** 6

**d.** 8

**6** Solve for $x \in [0, \pi]$: $\sqrt{3}\tan(2x) = 1$.

**a.** $\frac{\pi}{12}, \frac{7\pi}{12}$

**b.** $\frac{\pi}{6}, \frac{5\pi}{6}$

**c.** $\frac{\pi}{3}$

**d.** $\frac{\pi}{4}, \frac{3\pi}{4}$

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## Section 3: Inverse Trigonometric Functions

**7** What is the exact value of $\sin(\arccos(-\frac{1}{2}))$?

**a.** $\frac{\sqrt{3}}{2}$

**b.** $-\frac{\sqrt{3}}{2}$

**c.** $\frac{1}{2}$

**d.** $-\frac{1}{2}$

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**8** Evaluate $\tan(\arcsin(\frac{5}{13}))$.

**a.** $\frac{5}{12}$

**b.** $\frac{12}{5}$

**c.** $\frac{5}{13}$

**d.** $\frac{12}{13}$

**9** What is the range of the principal inverse tangent function $f(x) = \arctan(x)$?

**a.** $(-\frac{\pi}{2}, \frac{\pi}{2})$

**b.** $[-\frac{\pi}{2}, \frac{\pi}{2}]$

**c.** $[0, \pi]$

**d.** $(0, \pi)$

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## Section 4: Law of Tangents & Polar Trigonometry

**10** Which identity represents $\cos(A - B)$?

**a.** $\cos A \cos B + \sin A \sin B$

**b.** $\cos A \cos B - \sin A \sin B$

**c.** $\sin A \cos B - \cos A \sin B$

**d.** $\sin A \cos B + \cos A \sin B$

**11** In $\triangle ABC$, $a = 7$, $b = 5$, and $c = 8$. What is $\cos B$?

**a.** $\frac{11}{14}$

**b.** $\frac{1}{2}$

**c.** $\frac{5}{7}$

**d.** $\frac{13}{16}$

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**12** What is the maximum value of the function $f(x) = 3\sin x - 4\cos x$?

**a.** $1$

**b.** $5$

**c.** $7$

**d.** $\sqrt{7}$

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